

PH 512: Classical Mechanics

Course Code	PH 512
Course Name	Classical Mechanics
L-T-P-C	3-0-0-3
Prerequisites	Undergraduate Mechanics course
Intended for	Core for M.Sc. and I-Ph.D. in Physics; Elective for B.Tech. 3rd and 4th Year.
Mutual Exclusion	None
Approval	10th Senate (Jun 2016)
Revised	64th BoA (May 2026) - credits changed to 3-0-0-3

Preamble

Classical mechanics lies in the foundation of the mechanics of the observable world. Starting from Newtonian mechanics, this course looks into the conservation laws of mechanics, which helps to develop the analytical mechanics in terms of the generalized coordinates. The course looks into Lagrangian and Hamiltonian formalism and the framework of solvable models such as two-body, rigid-body, and small oscillations. Then, we look into the canonical transformation and associated algebraic framework required for the advanced formulation of the mechanics. This course is at an advanced level with more mathematical proofs that enhance the basic introduction to classical mechanics in the undergraduate level.

Course Modules

- **Unit/Topic 1: Introduction:** Mechanics of a system of particles, constraints, Lagrange's equation motion, Hamilton's principle and calculus of variations, Applications of the Lagrangian Formulation. Symmetries and conservation laws. (10 hours)
- **Unit/Topic 2: The Central Force Problem:** The Equivalent one-dimensional problem and classification of orbits, the Kepler problem. (6 Hours)
- **Unit/Topic 3: The Kinematics of Rigid Body motion:** Orthogonal transformations, Euler's theorem on the motion of a rigid body, finite rotations, infinitesimal rotations, rate of change of a vector, Angular momentum and kinetic energy of motion, the inertia tensor and the moment of inertia, Euler equation of motion of the rigid body. (8 Hours)
- **Unit/Topic 4: Oscillations:** Formulation of the problem, the eigenvalue equation and the principal axis transformation, small oscillations, frequencies of free vibration, and normal coordinates. (4 hours)
- **Unit/Topic 5: The Hamilton Equations of Motion:** Legendre Transformations and the Hamilton Equations of Motion, Cyclic Coordinates and Conservation Theorems, The Principle of Least Action. (6 Hours)
- **Unit/Topic 6: Canonical Transformations:** The examples of canonical transformation Poisson Bracket and canonical invariants, Liouville's theorem. Hamilton-Jacobi theory and Action-Angle Variables (8 Hours)

Laboratory/practical/tutorial Modules: None

Text Books

1. H. Goldstein, *Classical Mechanics*, 3e, Pearson Education; 3rd edition (2011), India.

References

1. C. Lanczos, *The variational principles of mechanics*, Dover Publications; New edition (1986), US.
2. N.C. Rana and P.S. Joag, *Classical Mechanics*, McGraw Hill Education, India.
3. L.D. Landau and E.M. Lifshitz, *Mechanics*; Third edition (1 January 2010), Elsevier, India.
4. John R Taylor *Classical Mechanics*, University Science Books (2005) US.